

Leveraging haplotype information in heritability estimation and polygenic prediction

Corresponding Author: Dr Jonas Meisner

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this article, Meisner et al has introduced a new reference free polygenic risk score method, utilizing haplotype information to improve both the heritability estimation and polygenic prediction. The paper is generally well written with well thought out experiments to demonstrate the power of the new method. However, we do have a few questions related to the simulation performed and the method as a whole:

1. Please correct us if we are wrong, our understanding of hapla is that it is an algorithm that tried to cluster the haplotypes within a genomic window and assigned the haplotype centroid as the representative "genotype" at the specific window. The haplotype matrix was then represented by an N times KW matrix where KW is the number of clusters within each window, with 1 denoting whether a sample is clustered into specific haplotype. Our first confusion was on how the genomic windows were defined. In the method section, it was stated that the genomic window can be overlapping or non-overlapping, was the window defined using pre-defined genomic regions? Or was a sliding window used (as illustrated in figure 1)? If a sliding window were used, how was the window size and step size defined? Were they by number of variants, or genomic distance?
2. Based on the algorithm, it seems like there are two hyper-parameters for hapla, the fix window size, and the lambda for delayed cluster creation. How were these parameters selected? In the application to iPsych data, was these hyper-parameters optimized in each cross validation?
3. It is not clear to us how the hapla can be applied to an independent dataset or how can the leave-one-out cross validation can be performed. Specifically, how would you assign the haplotype cluster to the leave of samples? Would the clustering be done w.r.t. the training sample, and then the leave-out / independent samples would be assigned to haplotype centroids that with the smallest Manhattan distance? If that is the case, would it mean that the haplotype centroids / the training samples would need to be shared with other researchers if they wish to apply the hapla model to their own dataset?
4. In the simulation, a 100Mb chromosomal segment were simulated. If we understand correctly, the phenotype was simulated using the down sampled genotype data, which contain no more than 50k SNPs. 1,000 causal variants were than randomly selected. While the simulated heritability of 0.2 and 0.8 might generally be reasonable in real life scenario, the relatively small number of variants might means lead to a higher than usual correlation between the "null" variants and the causal variants. This might have led to the over-estimation of heritability from GCTA, which is generally accurate with its heritability estimation in quantitative traits. An alternative simulation method might be to use existing large datasets such as iPSYCH or UKB to simulation phenotypes with more realistic distribution of variants.
5. An interesting, albeit complicated analyses that can be performed here is to first simulate the phenotype before doing the variant down-sampling. This would allow for the simulating the scenario where there is imperfect tagging.
6. We were surprised to see the correlation of the PGS from PRSice against the PCs. Were PCs included in the parameter optimization when using PRSice?
7. When compared to PRSice, were CV also performed for PRSice? Or were that only performed for hapla (or none?). The main concern here is the over-fitting of hapla to the iPsych data, which can lead to higher predictive performance.
8. One thing to note with the result interpretation is that PRSice generally is one of the worst performing PGS method. As such, one might be cautious to draw conclusion of hapla outperforming all reference-based polygenic risk score methods, especially when the performance of hapla is only slightly better than PRSice for schizophrenia. A more accurate presentation would be pinpointing the performance difference to that between hapla and PRSice only.
9. It is very helpful to include Figure S2 which shows the overall computation time required. However, it is of interest to also show the performance in real data, e.g. iPSYCH.

Minor comments:

1. Figure 4 – why PGC results were missing for REC?
2. In result section: “The heritability estimates of hapla were consistently always ...” can delete “always” as it is redundant.
3. In result section: “The relative increases in R2 for hapla in the full data were ..., respectively, and ..., when again compared to GCTA” can be re-written to “Compared to GCTA, the relative increases in R2 for hapla...” to allow for clarity.
4. Similarly, “The heritability estimates of hapla were slightly larger for the depression disorders but lower for schizophrenia” should add “Compared to GCTA”, as this is not true for the pruning method.

(Remarks on code availability)

The github seems to be documented and we would expect it to run, though we have not tested it.

Reviewer #2

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

Leveraging haplotype information in heritability estimation and polygenic prediction
Jonas Meisner et al.

This study introduces a framework named hapla, featuring a novel algorithm for clustering haplotypes in phased genotype data to estimate heritability and perform reference-free polygenic prediction in complex traits. The researchers used inferred haplotype clusters to compute accurate SNP heritability estimates and polygenic scores in both a simulation study and the iPSYCH2012 case-cohort for depression disorders and schizophrenia. The findings demonstrate that the proposed haplotype-based approach performs better than standard genotype-based approaches.

I have a number of questions and comments that should be considered to improve the current version of the manuscript.

1. In the overview of the hapla framework (Results), please clearly explain what is meant by a haplotype cluster and how it is distinguished from haplotyping. The study proposes a novel clustering method (PDP-DC-Medians), but it is not clear how haplotyping itself was performed (i.e., which method and software were used). It is important to clarify whether haplotype clustering is synonymous with haplotyping in this context. Providing a detailed explanation in this section will help avoid any confusion.
2. It is surprising that the study did not verify the accuracy of haplotyping (or haplotype clustering) in their simulations. What are the concordance and discordance rates for the alleles when comparing inferred haplotypes to true haplotypes across the genomic region? The same applies when comparing inferred clusters to true clusters in their simulations. Additionally, it is recommended to compare the performance of the proposed method with existing methods (there are many available for haplotyping and haplotype clustering).
3. It is unclear why the standard GREML method is biased for AFR and EUR when estimating SNP-based heritability (e.g., Figure 2). The authors should elaborate on how they simulated the phenotypes and the assumptions made in the simulations. Did they simulate phenotypes based on all data and then divide the data into AFR and EUR to estimate SNP-based heritability for each ancestry? Please clarify this in the relevant section and in the figure legends.
4. When comparing R^2 (Figures 2 and 3, and Table 1), do they assess the significance of the differences? Please clearly show the significance p-values for the differences. A comparative significance test can be done using the R package, *r2redux*. Refer to Momin et al., "Significance tests for R^2 of out-of-sample prediction using polygenic scores" (2023) *AJHG* 110: 349-358.
5. I wonder how the authors managed the haplotyping (clustering) effects in the prediction. Is there a possibility that batch effects in haplotyping and clustering might positively correlate with the accuracy of the prediction, potentially leading to inflation?
6. In Figure 3, it is unclear what target data were used. Did they employ 10-fold cross-validation? Please clarify this in the relevant section and the legend of the figures.
7. The authors should be aware that Nagelkerke R^2 is problematic. It has been quoted as follows:
“Nagelkerke’s R^2 (NR2) is often reported. However, the NR2 is difficult to benchmark since it depends on the proportion of cases in the target sample and is not on the liability scale, so it cannot be directly compared to heritability estimated from

twin or family data.”

Reference: Wray et al. Research Review: Polygenic methods and their application to psychiatric traits (2014) J Child Psychol Psychiatry 55:1068-87.

8. Figure 4 is unclear regarding its fairness in comparison. The training data used are different (leave-one-out iPSYCH data for hapla vs. PGC data for PGC). The prediction performance is influenced not only by the hapla method but also by other factors such as sample size, genetic correlation between the target and respective reference data, and the effective number of chromosome segments. This result may not directly reflect the superior performance of hapla. Therefore, what is the purpose of presenting this comparison here?

9. Please explain why the AFR population has a higher number of SNPs than the EUR population in the simulated data section of the Methods.

10. The phenotypic simulations section requires much greater detail. Were phenotypes simulated in the last generation for each population, or were they simulated in the founder generation and inherited through subsequent generations? How did the authors account for allele frequency differences between the two populations? What genetic correlation is assumed between the two populations? Do they assume that the same causal effects are shared between the two populations? If so, this assumption is highly unrealistic.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this update, the authors have significantly enhanced the clarity of the paper, and I appreciate their efforts. My only remaining comment is that they should include a discussion on the limitation of using only PRSice as a reference-based method. This choice may not fully represent the diversity of available reference-based methods.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The authors have addressed most of my concerns; however, a formal assessment of the significant difference between the proposed method and GCTA in the prediction context is still needed.

In the abstract, the authors claim, "We demonstrate that our haplotype-based approach robustly outperforms standard genotype-based approaches, which can help pave the way for polygenic risk scores in the future of precision medicine and psychiatry."

However, there is no clear evidence supporting that the haplotype-based method is statistically superior to the genotype-based approach (e.g., GCTA) in Figures 3 and 4. The 95% confidence intervals from the bootstrapping analysis largely overlap between the two methods, and do not convincingly show that the haplotype-based approach outperforms the genotype-based one.

There are several statistical approaches to properly assess the significance of the difference between these methods:

1. Paired t-test, as the authors used 10-fold cross-validation, which would allow comparison of performance across folds.
2. R2redux, which can be applied even to binary outcomes using a linear model to quantify improvement.
3. DeLong's test, which assesses the difference between AUCs (area under the curve) for binary classification models.

Implementing one or more of these methods would offer a more rigorous evaluation of the comparative performance, providing stronger evidence to substantiate their claim.

(Remarks on code availability)

Version 2:

Reviewer comments:

Reviewer #3

(Remarks to the Author)

The manuscript has been strengthened with clear statistical evidence supporting the conclusions, validated by significance testing. I have no further comments.

(Remarks on code availability)

Open Access This Peer Review File is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.

The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

Answers to reviewers

Leveraging haplotype information in heritability estimation and polygenic prediction

Comments and questions from the reviewers are colored black, while our answers are colored green.

Reviewer #1 and #2

In this article, Meisner et al has introduced a new reference free polygenic risk score method, utilizing haplotype information to improve both the heritability estimation and polygenic prediction. The paper is generally well written with well thought out experiments to demonstrate the power of the new method. However, we do have a few questions related to the simulation performed and the method as a whole:

Authors' reply: We thank the reviewers for their nice words and excellent questions with opportunities for clarifications that have improved our work substantially. We have addressed the comments in chronological order.

1. Please correct us if we are wrong, our understanding of hapla is that it is an algorithm that tried to cluster the haplotypes within a genomic window and assigned the haplotype centroid as the representative "genotype" at the specific window. The haplotype matrix was then represented by an N times KW matrix where KW is the number of clusters within each window, with 1 denoting whether a sample is clustered into specific haplotype. Our first confusion was on how the genomic windows were defined. In the method section, it was stated that the genomic window can be overlapping or non-overlapping, was the window defined using pre-defined genomic regions? Or was a sliding window used (as illustrated in figure 1)? If a sliding window were used, how was the window size and step size defined? Were they by number of variants, or genomic distance?

Authors' reply: We thank the reviewers for the comment, which is an excellent opportunity to clarify. Yes, the reviewers are correct that *hapla* clusters haplotype in genomic windows along the genome, which are then used to count the haplotype cluster assignments for each individual into a new "genotype" data structure. We agree that the overview of the *hapla* framework in the result section is too sparse with information regarding parameters and their definitions. We are defining genomic windows in a simple manner with a fixed size of B variants, which are overlapping. In our case, we are utilizing one overlapping window, such that the step-size is $\text{ceil}(B/2)$. Since different local haplotype architectures can be modeled in different window sizes (number of variants), we are therefore running *hapla* for multiple window sizes, $B = \{8, 16, 32\}$, independently, and estimating a GRM for each. We are hereafter combining the different haplotype GRMs in *GCTA*, including one based on SNPs alone ($B = 1$), for heritability estimation and polygenic prediction, which we refer to as the 'Multi' approach. We have added the following text to the "Overview of the hapla framework"

section in Results, and specified the step-size more clearly in the different analyses. We changed the subheading of “Window sizes in *hapla* and multiple genetic effects” to “Hyperparameters of *hapla* and multiple genetic effects” to emphasize that they are hyperparameters, including multiple minor edits.

“The size of the windows is determined as a fixed number of predefined SNPs, B , and windows are allowed to overlap, which effectively creates a sliding window approach for a given step-size, L , being $1 \leq L \leq B$ SNPs. Different choices of window sizes and step-sizes can model distinct aspects of the haplotype architecture.”, on page 3, paragraph 3.

“We inferred haplotype clusters in overlapping windows across the simulated chromosomal segment for multiple window sizes, $B = \{1, 8, 16, 32\}$ using a step-size $L = \lceil B/2 \rceil$ in each case, to capture potential recombination events (Table S2).”, on page 5, paragraph 2.

2. Based on the algorithm, it seems like there are two hyper-parameters for *hapla*, the fix window size, and the lambda for delayed cluster creation. How were these parameters selected? In the application to iPsych data, was these hyper-parameters optimized in each cross validation?

Authors' reply: We thank the reviewers for the comment. It is correct that there are two hyperparameters in the haplotype clustering of *hapla*, B and λ . As also mentioned in the previous comment, different window sizes for the haplotype clustering may capture different local architectures. We therefore tested different window sizes (1, 8, 16 and 32) in the simulation study to model distinct types of haplotype structure. We hereafter evaluated an approach combining the different GRMs for each of the fixed window sizes (multiple genetic effects) in the GREML approach of *GCTA* (`--mgrm`) to retain as much haplotype information as possible.

We chose 0.1 for λ as a simple choice, which means that 10% of the SNPs in a haplotype would have to be different compared to the already generated haplotype cluster medians. If this would be the case, a new haplotype cluster is generated based on that “extreme” haplotype. To demonstrate the sensitivity of *hapla* in the haplotype clustering and the subsequent heritability estimation and polygenic prediction, we have now included an extra simulation analysis, where we use $\lambda = 0.05$ and another for $L = B$, with non-overlapping windows. The lower λ value will generate more haplotype clusters in every window, for $B = \{16, 32\}$, while it will have no effect for $B = 8$, as $\lambda = 0.1$ or $\lambda = 0.05$ would check for 1 SNP differences in both cases. Here we showcase the robustness of *hapla* in terms of changing its hyperparameters.

We demonstrated in the simulation study that $\lambda = 0.1$ together with the ‘Multi’ approach were the best overall performing setup across all datasets, in comparison to using the haplotype cluster assignments of one window size, independently, so we decided to use the same set of hyperparameters for the iPSYCH2012 data. The iPSYCH2012 data has therefore not had any influence in the hyperparameter determination of *hapla*. To clarify this for the reader, we have inserted the following in the subsection “Hyperparameters of *hapla* and multiple genetic effects” of the Results section:

*“We tested the robustness of *hapla* in terms of choices for its hyperparameters (λ and L) using the Multi approach in Scenario 1 (Figure S8). The results were consistent across all*

three subsets for both the full and downsampled datasets, demonstrating that hapla is robust to hyperparameter changes, and we use the default values ($\lambda = 0.1$ and $L = \lceil B/2 \rceil$) in all following analyses.”, page 5, paragraph 4.

3. It is not clear to us how the hapla can be applied to an independent dataset or how can the leave-one-out cross validation can be performed. Specifically, how would you assign the haplotype cluster to the leave of samples? Would the clustering be done w.r.t. the training sample, and then the leave-out / independent samples would be assigned to haplotype centroids that with the smallest Manhattan distance? If that is the case, would it mean that the haplotype centroids / the training samples would need to be shared with other researchers if they wish to apply the hapla model to their own dataset?

Authors' reply: We thank the reviewers for the comment and it will be important for future applications. In the *hapla* software, we have a simple flag “--medians” in the haplotype clustering component (*hapla cluster*), which allows one to save the haplotype cluster medians for all evaluated windows. Then as you correctly assume, one would assign new haplotypes to clusters based on Manhattan distance. We have already implemented this as a haplotype cluster prediction component, *hapla predict*, in our framework, which further allows for missingness in the new independent haplotypes. The haplotype cluster medians are contained in a lightweight data structure, in terms of storage, such that we envision researchers to share them in a similar manner to GWAS summary statistics. As haplotype cluster medians are not representing individual data, we expect sharing of cluster medians to be easy in larger consortiums with strict data policies, but we deem it out of scope for further discussion in this method paper. We have now provided more details about how one would go about obtaining haplotype cluster assignments in an independent dataset, at the end of “Clustering algorithm” and “Implementation details” in the Methods section:

“The haplotype cluster medians can further be used to infer cluster assignments of unseen haplotypes on the same set of variants. The unseen haplotypes will be assigned to the cluster median with the smallest Manhattan distance, allowing for some degree of missingness.”, page 14, paragraph 2.

“The haplotype cluster medians can be saved as a binary array collection, such that they can easily be utilized to cluster unseen haplotypes of an independent phased genotype dataset.”, page 16, paragraph 3.

4. In the simulation, a 100Mb chromosomal segment were simulated. If we understand correctly, the phenotype was simulated using the down sampled genotype data, which contain no more than 50k SNPs. 1,000 causal variants were than randomly selected. While the simulated heritability of 0.2 and 0.8 might generally be reasonable in real life scenario, the relatively small number of variants might means lead to a higher than usual correlation between the “null” variants and the causal variants. This might have led to the over-estimation of heritability from GCTA, which is generally accurate with its heritability estimation in quantitative traits. An alternative

simulation method might be to use existing large datasets such as iPSYCH or UKB to simulate phenotypes with more realistic distribution of variants.

5. An interesting, albeit complicated analyses that can be performed here is to first simulate the phenotype before doing the variant down-sampling. This would allow for the simulating the scenario where there is imperfect tagging.

Authors' reply: We thank the reviewers for the excellent questions and suggestions, and we have combined question 4 and 5, so they can be addressed and clarified together. We have only simulated phenotypes based on the full SNP sets, so we apologize for the confusion as we have done as the reviewers suggested in their fifth comment. Our idea with the downsampled genotype datasets is exactly what the reviewer described, to evaluate the performance of the different methods when having limited SNP information and imperfect tagging, thus the phenotypes remain the same and are only defined from the full SNP set. We have now updated and added text to further clarify that phenotypes are only generated from the full SNP set in the beginning of the subsection "Heritability estimation and polygenic prediction in simulation study" in the Results section and in the "Phenotype simulations" subsection of the Methods section.

"We simulated phenotypes independently in each population for three different scenarios, sampling 1000 causal loci from the full variant sets, replicated 10 times. Scenario 1 and 2 were SNP-based assuming $h^2 = 0.8$ and $h^2 = 0.2$, respectively, and Scenario 3 was haplotype cluster-based assuming $h^2 = 0.8$ using overlapping windows of $B=8$ SNPs.", page 4, paragraph 1.

"We follow the simple approach in [32], where we randomly sample $M_c = 1,000$ causal variants from the full SNP sets.", page 17, paragraph 2.

6. We were surprised to see the correlation of the PGS from PRSice against the PCs. Were PCs included in the parameter optimization when using PRSice?

Authors' reply: We thank the reviewers for the comment and it is a very good point. Our analyses with PRSice were performed without covariates. We have now *updated* all results regarding PRSice to include covariates in its parameter optimization, leading to new models as other optimal thresholds of course have been found. However, this also means that there has been an overall performance degradation for their estimated polygenic risk scores, in both DEP and SCZ. To our surprise, the polygenic risk scores from PRSice are still highly correlated with the principal components showing that the inclusion of covariates did not mitigate the issue, when optimizing PRSice. The correlation with the top principal components significantly increased in the best models of PRSice. This further indicates that PRSice (and other reference-based approaches) also models non-genetic effects. Its best model for SCZ is now the most performant in the case of individuals with Danish born parents, however, it has an even bigger performance degradation for the full sample set or for individuals with one parent not being Danish. We have updated Figure S9 with the new correlation values along with the subsection "Reference-based polygenic risk scores from summary statistics" describing these issues in the Results section, and updated our discussion with the following:

“We further computed correlations between the reference-based polygenic risk scores and the estimated principal components, which were used as fixed-effects covariates in all the reference-free estimations and in the model optimization of PRSice, for indications of uncorrected population structure in the original GWASs.”, page 9, paragraph 2.

“The well known portability problem affected the reference-based approach to a very large degree, such that it lost most of its predictive performance in MDD and SCZ.”, page 11, paragraph 2.

7. When compared to PRSice, were CV also performed for PRSice? Or were that only performed for hapla (or none?). The main concern here is the over-fitting of hapla to the iPsych data, which can lead to higher predictive performance.

Authors' reply: Thank you for your comment. We ran PRSice with default parameters such that it optimizes R^2 in the iPSYCH2012 case-cohort directly, and we have only reported the best model of PRSice. You could therefore argue that PRSice is overfitted to the iPSYCH2012 case-cohort further showcasing its poorer performance in comparison to reference-free estimators in this case. We have now added further information regarding the optimal choices found by PRSice in the two different scenarios (DEP and SCZ) in the Results section (now including covariates as suggested in previous comment of the reviewers), which are also described in Table S11, and extended texts regarding the optimization process of PRSice in the Results section under “Reference-based polygenic risk scores from summary statistics” and the Methods section under “Polygenic prediction”, with the following:

“The reference-based polygenic risk scores were based on ~2.5x and ~20x more cases than were available in the iPSYCH2012 case-cohort for MDD and SCZ, respectively, and the best models optimizing R^2_N were reported (Table S11).”, page 9, paragraph 1.

“We use PRSice (v2.3.5) [10] to estimate polygenic risk scores based on a clumping and thresholding (C+T) approach using its default parameters that optimizes for R^2 directly in the target dataset.”, page 15, paragraph 8.

8. One thing to note with the result interpretation is that PRSice generally is one of the worst performing PGS method. As such, one might be cautious to draw conclusion of hapla outperforming all reference-based polygenic risk score methods, especially when the performance of hapla is only slightly better than PRSice for schizophrenia. A more accurate presentation would be pinpointing the performance difference to that between hapla and PRSice only.

Authors' reply: We thank the reviewers for the comment. We agree that PRSice is only one single software out of many different reference-based polygenic risk score estimators. However, PRSice is still the most widely used polygenic risk score estimation software, as it also has much fewer assumptions than other tools. Most other reference-based approaches require the use of only a subset of variants (usually HapMap3) due to the extra reliance of a LD map, which they use to further adjust GWAS summary statistics. We have updated and

added text in the Results and Discussion sections to reflect that we are outperforming PRSice rather than all reference-based estimators in general.

“We computed reference-based polygenic risk scores using PRSice, a standard clumping and thresholding approach and one of the most widely used software for polygenic risk score estimation. We used GWAS summary statistics from individuals of European ancestry in the Psychiatric GWAS Consortium (PGC) [15,54] excluding iPSYCH case-cohorts for both MDD and SCZ and excluding 23andMe samples for MDD to evaluate it against the reference-free approaches.”, page 9, paragraph 1.

“This was in stark contrast to the reference-based polygenic risk scores of PRSice that collapsed with relative decreases of 85.2% and 85.7% for MDD and SCZ, respectively.”, page 10, paragraph 1.

“However, we further investigated the correlation between the reference-based polygenic risk scores of PRSice and the estimated principal components in the iPSYCH2012 case-cohort, as it is known to potentially inflate their predictive performance, which was the case for PRSice in both MDD and SCZ.”, page 11, paragraph 2.

“We further note that polygenic risk scores from the widely used PRSice, based on an optimized clumping and thresholding approach, may not represent all reference-based approaches, and other methods could have better predictive performance. However, most other reference-based approaches rely on population-specific LD maps, heavily restricting the SNP set used, which could deteriorate performance in diverse populations even further.”, page 12, paragraph 2.

9. It is very helpful to include Figure S2 which shows the overall computation time required. However, it is of interest to also show the performance in real data, e.g. iPSYCH.

Authors' reply: We thank the reviewers for the comment and we fully agree. Since first submission of the manuscript, we have implemented computational updates to our clustering algorithm (v0.11), which has provided major speed-ups in the runtime. This also changed how we deal with ties between haplotype cluster distances, such that some results have the slightest deviation from the first submission. We have now provided computational runtimes for chromosome 1 for the datasets in the iPSYCH2012 case-cohort, as well in Figure S2. We have added the following text to the subsections “Heritability estimation and polygenic prediction in simulation study” and “Heritability estimation and reference-free polygenic risk scores” in the Results:

“We clustered the haplotypes in less than 3.5 minutes across all window sizes for all datasets, showcasing the speed of our clustering algorithm (Figure S2).”, page 5, paragraph 2.

“We clustered the haplotypes of chromosome 1 in less than 9 minutes for all three mental disorders using hapla, demonstrating again its fast functionalities (Figure S2).”, page 8, paragraph 1.

Minor comments

1. Figure 4 – why PGC results were missing for REC?

Authors' reply: Unfortunately, the PGC does not provide GWAS summary statistics on the subtypes of depression such as recurrent depressive disorder (REC). We have added the following sentence to the subsection, 'Reference-based polygenic risk scores from summary statistics', and to the captions of Figure 4 and table S11 and S12 for further clarification:

"PGC GWAS summary statistics were not available for REC."

2. In result section: "The heritability estimates of hapla were consistently always ..." can delete "always" as it is redundant.
3. In result section: "The relative increases in R2 for hapla in the full data were ..., respectively, and, when again compared to GCTA" can be re-written to "Compared to GCTA, the relative increases in R2 for hapla..." to allow for clarity.
4. Similarly, "The heritability estimates of hapla were slightly larger for the depression disorders but lower for schizophrenia" should add "Compared to GCTA", as this is not true for the pruning method.

Authors' reply: We thank the reviewers for the very nice suggestions. We have incorporated them all into our revised manuscript.

Reviewer #3

This study introduces a framework named hapla, featuring a novel algorithm for clustering haplotypes in phased genotype data to estimate heritability and perform reference-free polygenic prediction in complex traits. The researchers used inferred haplotype clusters to compute accurate SNP heritability estimates and polygenic scores in both a simulation study and the iPSYCH2012 case-cohort for depression disorders and schizophrenia. The findings demonstrate that the proposed haplotype-based approach performs better than standard genotype-based approaches.

I have a number of questions and comments that should be considered to improve the current version of the manuscript.

Authors' reply: We thank the reviewer for taking their time and their questions and comments, which have certainly improved our manuscript. We have addressed them in a chronological order.

1. In the overview of the hapla framework (Results), please clearly explain what is meant by a haplotype cluster and how it is distinguished from haplotyping. The study proposes a novel clustering method (PDP-DC-Medians), but it is not clear how haplotyping itself was performed (i.e., which method and software were used). It is

important to clarify whether haplotype clustering is synonymous with haplotyping in this context. Providing a detailed explanation in this section will help avoid any confusion.

Authors' reply: We thank the reviewer for the comment and the opportunity to clarify. Our framework requires phased genotype data such that haplotype estimation (phasing/haplotyping) has already been performed. It is therefore not a new algorithm for haplotyping, but the proposed haplotype clustering algorithm is a window-based process that is performed after the haplotyping (phasing). Our haplotype clustering is a very scalable approach for condensing and allowing haplotype information to be used in downstream analyses, such as heritability estimation and polygenic prediction here in this study. Haplotype information can unfortunately not be used directly in subsequent analyses due to switch errors and arbitrary ordering of haplotypes across chromosomes. We mitigate these issues by constructing and aggregating our window-based haplotype cluster alleles such that we can model individuals.

For the iPSYCH2012 case-cohort, we utilize a phased and imputed version which has already been thoroughly evaluated and we have therefore not performed haplotype estimation ourselves in this study (<https://doi.org/10.1038/s42003-023-04477-y>). The phasing and imputation was reference-based using the Haplotype Reference Consortium v1.1 (64,976 haplotypes) as reference panel. We have added more details to the "Overview of the *hapla* framework" in the Results section emphasizing that the haplotype clustering is performed in windows from phased genotype data (after haplotyping), and added a clearer description of a haplotype cluster for clarification and to avoid confusion.

"Haplotype clusters are modeled as cluster medians that act as representatives of all their assigned haplotypes. We therefore assign haplotypes to the inferred clusters based on the Manhattan distance between the corresponding cluster median and the haplotype, to obtain window-based haplotype cluster assignments across the genome for each individual.", page 3, paragraph 3.

2. It is surprising that the study did not verify the accuracy of haplotyping (or haplotype clustering) in their simulations. What are the concordance and discordance rates for the alleles when comparing inferred haplotypes to true haplotypes across the genomic region? The same applies when comparing inferred clusters to true clusters in their simulations. Additionally, it is recommended to compare the performance of the proposed method with existing methods (there are many available for haplotyping and haplotype clustering).

Authors' reply: We thank the reviewer for the comment. We have assumed that this comment relates to the previous comment of the reviewer, and we want to highlight that haplotype clustering is different from haplotyping (haplotype estimation/phasing). In the simulations, the *msprime* program samples haploid chromosomes in the last generation (present time) resembling whole-genome sequencing with no errors, and it simply combines two sampled haploid chromosomes to create diploid individuals. The diploid individuals are saved into phased BCF/VCF files. We are therefore only working directly on the ground-truth haplotypes throughout the simulations, and there is no need for *post-hoc* haplotype estimation. As mentioned in the previous comment, we are utilizing an already phased and

imputed genotype dataset of the iPSYCH2012 case-cohort, which has undergone extensive testing (<https://doi.org/10.1038/s42003-023-04477-y>). We have added further details in the Methods section under “Simulated data”:

“msprime simulates entire chromosomal segments such that the phasing information is known in the assembly of all diploid individuals.”, page 16, paragraph 4.

3. It is unclear why the standard GREML method is biased for AFR and EUR when estimating SNP-based heritability (e.g., Figure 2). The authors should elaborate on how they simulated the phenotypes and the assumptions made in the simulations. Did they simulate phenotypes based on all data and then divide the data into AFR and EUR to estimate SNP-based heritability for each ancestry? Please clarify this in the relevant section and in the figure legends.

Authors' reply: We thank the reviewer for the comment and question. It has been common in multiple software papers to simulate phenotypes from heritability models that may benefit a specific method. This point has been further discussed in “Reevaluation of SNP heritability in complex human traits”, by Doug Speed et al., *Nature Genetics* (<https://doi.org/10.1038/ng.3865>). We have therefore simulated phenotypes in a simple but efficient approach with minimal assumptions. Here the 1000 causal effects are sampled from a Gaussian distribution, $N(0, h^2/1000)$, and hereafter simply scaled to ensure that the variance of the genetic contribution is exactly h^2 . We are simulating phenotypes independently in each of the three datasets (AFR, EUR and ALL), which we also touch upon in their last comment. We have clarified the assumptions in the Results section, when describing the simulation study:

“We simulated phenotypes independently in each population for three different scenarios, sampling 1000 causal loci from the full variant sets, replicated 10 times. The genetic liabilities were defined as an additive model, and the causal effects were assumed to be Gaussian.”, page 4, paragraph 1.

4. When comparing R^2 (Figures 2 and 3, and Table 1), do they assess the significance of the differences? Please clearly show the significance p-values for the differences. A comparative significance test can be done using the R package, r2redux. Refer to Momin et al., “Significance tests for R^2 of out-of-sample prediction using polygenic scores” (2023) *AJHG* 110: 349-358.

Authors' reply: We thank the reviewer for the comment. In the simulation study, we have replicated each scenario 10 times, such that the error bars in the results of Figure 2 are displaying the mean and standard deviations. It is therefore not suitable for significance testing, as it is displaying the results of 10 different simulated phenotypes.

The r2redux R package appears to only be using the standard measure for coefficient of determination, R^2 , derived from “ $\text{lm}(y \sim \text{PGS})$ ”, which doesn't hold for binary responses but only continuous traits. We therefore do not expect the theoretical basis of the significance test, based on the linear model on the observed scale, to hold for the binary

responses in the iPSYCH2012 case-cohort. Our reported measures R^2_N and R^2_L for the predictive performances are also not included for testing in the proposed *r2redux R* package.

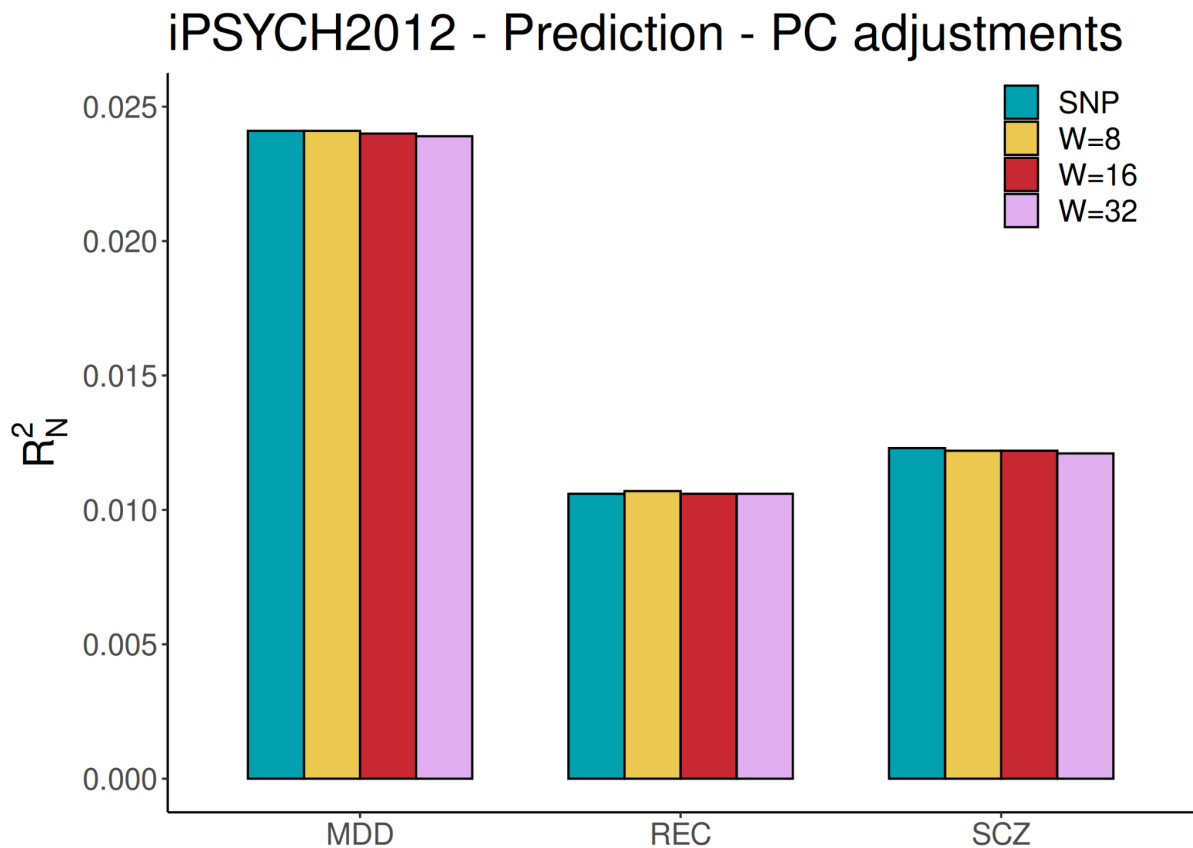
We have now added valid prediction intervals using bootstrapping in Figure 3 and 4 on the polygenic predictions, which is based on 1000 iterations to give the reader a better sense of the variability in the predictions. The following text has been added to their captions:

“The error bars of the heritability estimates represent the 95% confidence interval, while the error bars of the polygenic predictions represent the 95% prediction interval based on 1000 bootstrap iterations.”, caption Figure 3.

“The error bars represent the 95% prediction interval for the polygenic predictions based on 1000 bootstrap iterations.”, caption Figure 4.

5. I wonder how the authors managed the haplotyping (clustering) effects in the prediction. Is there a possibility that batch effects in haplotyping and clustering might positively correlate with the accuracy of the prediction, potentially leading to inflation?

Authors' reply: We thank the reviewer for the question. In the iPSYCH2012 case-cohort, we adjusted for age, gender and the top 20 principal components after quality control with the removal of variants and related individuals. The principal components will account for population structure and potential batch effects there may persist in the data. The principal components are derived from the phased and imputed data, such that batch effects due to haplotyping would also affect the imputation process, and be captured modeled by the principal components that are used for correction in the mixed model. We therefore expect the haplotype clusters, a derived data structure from the phased and imputed genotype data, to only model genetic information. The amount and frequencies of the generated haplotype cluster will of course be dataset-specific just like allele frequencies as an example, and we only include haplotype clusters which have a frequency of 0.01, through iterative re-clustering as described in the Methods section. To demonstrate this point, we here showcase that our *hapla* results stay consistent, in terms of predictive performance, while correcting for principal components derived from the haplotype cluster assignments in the estimation of cvBLUPs. Here the principal components based on our haplotype cluster assignments will correct for possible batch effects related to them. We show that for all three window sizes ($B = \{8, 16, 32\}$), and the original SNP-based principal components, in the following figure:



6. In Figure 3, it is unclear what target data were used. Did they employ 10-fold cross-validation? Please clarify this in the relevant section and the legend of the figures.

Authors' reply: We thank the reviewer for the comment. We have employed the leave-one-out cross validated polygenic predictions (cvBLUPs) throughout our paper for reference-free polygenic risk score estimation, <https://doi.org/10.1089%2Fcmb.2019.0325>. There is therefore no defined "target data" in the sense that all other individuals are used for every single individual (one vs. all) to estimate the in-sample polygenic risk scores, thus avoiding overfitting as showcased in their original paper. We have added more sentences highlighting the usage of the leave-one-out cross-validated polygenic predictions in the results section and the captions of figure 2 and 3 for clarification, as well as supplementary figure captions mentioning cvBLUPs.

In "Overview of the *hapla* framework" of the Results section:

"We finally use the estimated GRM based on haplotype cluster alleles in the GCTA [62] software to estimate SNP heritability and leave-one-out cross-validated polygenic predictions (cvBLUPs) in a linear mixed model (Step 3, Figure 1).", page 3, paragraph 3.

"Top row shows the SNP heritability estimates using the different GRMs across three subsets using either full or downsampled data, while the bottom row are the squared correlations between the leave-one-out cross-validated polygenic predictions (cvBLUPs) and the simulated phenotypes.", caption Figure 2.

“Left plot shows the SNP heritability estimates on the liability scale using the different GRMs across the three disorders, while the right plot shows the Nagelkerke's pseudo R^2 between the leave-one-out cross-validated polygenic predictions (cvBLUPs) and the binary outcomes.”, caption Figure 3.

7. The authors should be aware that Nagelkerke R^2 is problematic. It has been quoted as follows: “Nagelkerke's R^2 (NR2) is often reported. However, the NR2 is difficult to benchmark since it depends on the proportion of cases in the target sample and is not on the liability scale, so it cannot be directly compared to heritability estimated from twin or family data.” Reference: Wray et al. Research Review: Polygenic methods and their application to psychiatric traits (2014) J Child Psychol Psychiatry 55:1068-87.

Authors' reply: We thank the reviewer for the comment, and it is an excellent point. For all our heritability estimates in the iPSYCH2012 case-cohort, we report them on both the observed and the liability scale, where the liability scale relies on a specified population prevalence that may vary from study to study. However, as the reviewer also remarks, Nagelkerke's pseudo R^2 is still one of the most commonly used metrics, which is also why we reported it for the iPSYCH2012 case-cohort for comparisons between the approaches in our study using the same data. We have now added the Lee's pseudo R^2 in Table 1 and S11, including mentioning it in their respective captions, enabling easier future comparisons as you describe. We also added a new supplementary table (S13), re-creating S12, with Lee's pseudo R^2 instead of Nagelkerke's R^2 . Lee's pseudo R^2 is more interpretable on the liability scale with case-control ascertainment bias (<https://doi.org/10.1002/gepi.21614>). We have added the following text to the Results section under “Predicting mental disorders in the iPSYCH2012 case-cohort”:

“The polygenic risk scores were evaluated using both Nagelkerke's pseudo R^2_N and Lee's pseudo R^2_L metrics, where Lee's pseudo R^2_L is more interpretable on the liability scale and takes into account case-control ascertainment bias [23].”, page 7, paragraph 4.

8. Figure 4 is unclear regarding its fairness in comparison. The training data used are different (leave-one-out iPSYCH data for hapla vs. PGC data for PGC). The prediction performance is influenced not only by the hapla method but also by other factors such as sample size, genetic correlation between the target and respective reference data, and the effective number of chromosome segments. This result may not directly reflect the superior performance of hapla. Therefore, what is the purpose of presenting this comparison here?

Authors' reply: We thank the reviewer for the comment and question. The reason why we have chosen to include a reference-based polygenic risk score estimator is simply because it is the most commonly used approach in precision medicine and (human) genomic prediction. The reference summary statistics, chosen for our study, are from the largest GWAS consortium (PGC), which are based on far larger sample sizes than in our study. As we mention in the introduction, then recent work even shows the combination of

reference-free and reference-based estimators can lead to an increased predictive performance (<https://doi.org/10.1016/j.ajhg.2021.04.014>), so the two approaches are evaluated together. The large GWASs in our study are only based on European individuals and are therefore highly relevant, as they are the best possible scenario for a reference-based approach, minimizing portability issues. We agree that it is not directly comparable due to the factors the reviewer also mentions, however, reference-based estimators often have better predictive performance than reference-free, which we also see with the updated iPSYCH2012 results for SCZ in individuals with both Danish parents (updated results). We also wanted with our study to emphasize that reference-based estimators may not be the best choice (in larger datasets with individual-level information), thus we showcase the stark differences in predictive performance for genomic prediction. We further highlight issues regarding reference-based approaches regarding portability (related to the genetic correlation, of course), and the inadequate adjustment of confounders in the original GWASs and how it may affect predictive performances. We have tried to address and discuss these topics more clearly in the Results section under “Reference-based polygenic risk scores from summary statistics” and in the Discussion:

“We computed reference-based polygenic risk scores using PRSice, a standard clumping and thresholding approach and one of the most widely used software for polygenic risk score estimation. We used GWAS summary statistics from individuals of European ancestry in the Psychiatric GWAS Consortium (PGC) [15,54] excluding iPSYCH case-cohorts for both MDD and SCZ and excluding 23andMe samples for MDD to evaluate it against the reference-free approaches.”, page 9, paragraph 1.

“We further compared hapla to a widely-used reference-based approach (PRSice) using state-of-the-art GWAS summary statistics from European samples of the Psychiatric GWAS Consortium (PGC) for MDD and SCZ.”, page 11, paragraph 2.

9. Please explain why the AFR population has a higher number of SNPs than the EUR population in the simulated data section of the Methods.

Authors’ reply: We thank the reviewer for the comment. Our simulations are based on a ‘Two population out-of-Africa’ demographic model in *stdpopsim*. The demographic model includes a bottleneck event that occurred in the EUR branch during the out of Africa event. Here a large proportion of genetic diversity is lost for the EUR branch leading to a lower number of SNPs than in comparison to the AFR branch, as we also see in real modern human populations. We have added a sentence to clarify the reason why the AFR population has a higher number of SNPs. We have added the following text to the “Simulated data” subsection of the Methods for clarification:

“The AFR subset has a larger number of SNPs, reflecting the increased genetic diversity in comparison to the EUR subset, as the EUR population underwent a severe bottleneck in the out-of-Africa event as defined in the demographic model.”, page 16, paragraph 4.

10. The phenotypic simulations section requires much greater detail. Were phenotypes simulated in the last generation for each population, or were they simulated in the

founder generation and inherited through subsequent generations? How did the authors account for allele frequency differences between the two populations? What genetic correlation is assumed between the two populations? Do they assume that the same causal effects are shared between the two populations? If so, this assumption is highly unrealistic.

Authors' reply: We thank the reviewer for the comment and questions. The phenotypes have all been simulated in the last generation (present time) of a population. Throughout the simulation study, we are evaluating the performance of the methods in three different datasets, an African population (AFR), an European population (EUR) and a simple mixed population (ALL, no migration or admixture), where the two other populations are merged. We have simply simulated phenotypes in each of the three populations, independently, but we agree that this might be unrealistic for the mixed population if one is assuming different population causal effects, as you mentioned. However, we wanted to keep the simulation study consistent and easily reproducible. To ensure that population differences are not modeled improperly, for the mixed population (ALL), we have now added estimated principal components in all simulation analyses, as well as for AFR and EUR. They have been estimated independently in each dataset, to be as consistent as possible in regards to the procedure for real data with the iPSYCH2012 case-cohort. We have therefore *updated all* simulation results for heritability estimation and polygenic prediction. We have added the following text to "Heritability estimation and polygenic prediction in simulation study" in the Results section and "Phenotype simulations" in the Methods section for further clarification:

"We included the top 20 principal components as fixed-effects covariates to account for population structure in the genome-based restricted maximum likelihood (GREML) and cvBLUP estimations, which were estimated independently for each dataset.", page 5, paragraph 1.

"Phenotypes are simulated independently in each generated dataset to ensure correct scaling of the genetic liabilities and environmental noise.", page 17, paragraph 4.

Answers to reviewers

Leveraging haplotype information in heritability estimation and polygenic prediction

Comments and questions from the reviewers are colored black, while our answers are colored green.

Reviewer #1 and #2

In this update, the authors have significantly enhanced the clarity of the paper, and I appreciate their efforts. My only remaining comment is that they should include a discussion on the limitation of using only PRSice as a reference-based method. This choice may not fully represent the diversity of available reference-based methods.

Authors' reply: We thank the reviewers for the kind words and their excellent comment. We have now expanded on their point in our discussion:

"We further note that the reference-based polygenic risk scores from the widely used PRSice, leveraging an optimized clumping and thresholding approach, may not represent the broader range of reference-based approaches, and other available methods could achieve better predictive performance. However, most other reference-based approaches rely on population-specific LD maps, heavily restricting the SNP set used, which could also deteriorate performance in diverse populations even further, or would require the need for acquiring summary statistics from multiple large-scale GWAS, covering a diverse range of populations [47].", page 12, paragraph 2.

Reviewer #3

The authors have addressed most of my concerns; however, a formal assessment of the significant difference between the proposed method and GCTA in the prediction context is still needed.

In the abstract, the authors claim, "We demonstrate that our haplotype-based approach robustly outperforms standard genotype-based approaches, which can help pave the way for polygenic risk scores in the future of precision medicine and psychiatry."

However, there is no clear evidence supporting that the haplotype-based method is statistically superior to the genotype-based approach (e.g., GCTA) in Figures 3 and 4. The 95% confidence intervals from the bootstrapping analysis largely overlap between the two methods, and do not convincingly show that the haplotype-based approach outperforms the genotype-based one.

There are several statistical approaches to properly assess the significance of the difference between these methods:

1. Paired t-test, as the authors used 10-fold cross-validation, which would allow comparison of performance across folds.
2. R2redux, which can be applied even to binary outcomes using a linear model to quantify improvement.
3. DeLong's test, which assesses the difference between AUCs (area under the curve) for binary classification models.

Implementing one or more of these methods would offer a more rigorous evaluation of the comparative performance, providing stronger evidence to substantiate their claim.

Authors' reply: We thank the reviewer for the great suggestion to provide stronger evidence for *hapla* outperforming the other approaches. We have now performed statistical tests in the analyses of the iPSYCH2012 case-cohort using the *r2redux* R package as suggested by the reviewer and reported the statistically significant tests ($\alpha = 0.05$) for differences in R^2 between *hapla* and mainly GCTA and the reference-based polygenic risk scores, with the below additions to the manuscript:

"We have further performed statistical significance tests on the difference in R^2 metrics using the r2redux R package based on linear models [36].", page 7, paragraph 4.

"We found that the differences in R^2 between hapla and the two approaches are statistical significant for all evaluated cases using r2redux, with $p = 0.0072$, $p = 0.019$ and $p = 0.031$, when evaluated against the standard approach in GCTA, and $p = 1.2 \times 10^{-6}$, $p = 0.0015$ and $p = 0.0076$, evaluated against the LD pruning approach for MDD, REC and SCZ, respectively.", page 8, paragraph 1.

"The difference in R^2 between hapla and the reference-based polygenic risk scores was statistically significant using r2redux for MDD ($p = 1.2 \times 10^{-32}$).", page 9, paragraph 1.

"We found the difference in R^2 using r2redux between hapla and the standard genotype-based approach in GCTA statistically significant for MDD in the subset of having at least one non-Danish born parent ($p = 0.046$). There were still statistical significant differences for MDD between hapla and the reference-based polygenic risk scores in both subsets ($p = 5.4 \times 10^{-14}$ and $p = 6.0 \times 10^{-6}$), as well as for SCZ, in the subset of having at least one non-Danish born parent ($p = 0.042$).", page 10, paragraph 2.